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④ Teste

Questão ①

0,100 mol — Um gás monoatômico.

$P_1 = 1,00 \times 10^5 Pa$ — $C_V = \frac{3}{2}R$ e $C_P = \frac{5}{2}R$

$V_1 = 2,50 \times 10^{-3} \cancel{Pa} m^3$

a) $T_1$ em kelvins.

$$PV = nRT \Rightarrow T = \frac{PV}{nR} \quad T_1 = \frac{P_1 V_1}{nR}$$

$$T_1 = \frac{(1,00 \times 10^5 Pa)(2,50 \times 10^{-3}) \cancel{Pa}\, m^3}{(0,100\, mol)(8.31 \frac{J}{mol\,K})}$$

$$= \frac{250\, (N/m^2) \cdot m^3}{0,831 \frac{N \cdot m}{K}} \approx 300,84\, K$$

A temperatura Inicial foi de 300,84 K.

b) Após uma expansão de $V_i \rightarrow 2V_i$, encontre $T_f$ e $P_f$ para as expansões:

1) Isotérmica:

$$\Delta T = 0 \quad T_f = T_i \quad T_f = 300,84\, K.$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow P_1 V_1 = P_2 V_2 \Rightarrow P_1 V_1 = 2 P_2 V_1$$

$$P_1 = 2P_2 \qquad P_2 = \frac{P_1}{2} \qquad P_2 = \frac{1,00 \times 10^5 Pa}{2}$$

$$P_2 = 5,00 \times 10^4 Pa$$

ii) Isobárico

$\Delta P = 0 \quad P_f = P_i \quad P_f = 1{,}00 \times 10^5\ Pa$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \qquad \frac{V_1}{T_1} = \frac{V_2}{T_2} \qquad T_2 = T_1 \frac{V_2}{V_1}$$

$$T_2 = (300{,}84\,K)\frac{(2V_1)}{V_1} = 601{,}684\ K //$$

iii) Adiabático.

No processo adiabático. $\Delta Q = 0$

$\Delta U = Q - W \quad \Delta U = -W$

Se o gás expandiu, $W > 0 \Rightarrow \Delta U < 0$

No processo adiabático, o produto

$PV^{\gamma} = \text{const}$ e $TV^{(\gamma-1)} = \text{const.}$, $\gamma = \frac{C_P}{C_V}$

$$P_1 V_1^{\gamma} = P_2 V_2^{\gamma} = P_2 (2V_1)^{\gamma} = \cancel{P_2}$$

$$P_1 = P_2 \cdot 2^{\gamma} \quad P_2 = P_1 \cdot 2^{-\gamma}$$

$C_P = 5/2\,R \quad C_V = 3/2\,R \quad \frac{C_P}{C_V} = \frac{5/2\,R}{3/2\,R} = 5/3 //$

$P_2 = (1{,}00 \times 10^5\,Pa) \cdot 2^{-\frac{5}{3}} \quad P_2 \approx 31498{,}02\ Pa$

$$T_1 V_1^{(\gamma-1)} = T_2 (2V_1)^{(\gamma-1)}$$

$$T_2 = T_1 \left(\frac{V_1}{2V_1}\right)^{(\gamma-1)} = \frac{T_1}{2^{(\gamma-1)}}$$

$$= \cancel{\frac{300,84K}{2^{(\frac{5}{2}-1)}}} = \cancel{\frac{300,34K}{2^{(3/2)}}} = \frac{300,84K}{2^{(\frac{5}{3}-1)}}$$

$$= \frac{300,84K}{2^{\frac{2}{3}}} = 189,51\,K$$

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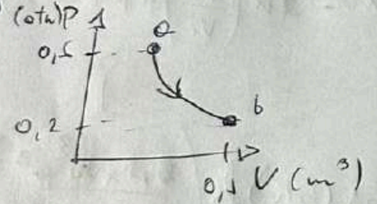
Questão 2

a) nº de molos.

$T = 85°C = 358,15\,K$

$PV = nRT$

$n = \frac{PV}{RT}$




Aplicando no ponto b: b: $\begin{cases} T_b = 85°C = 358,15K \\ P_b = 0,2\,atm = 20265\,Pa \\ V_b = 0,1\,m^3 \end{cases}$

$$n = \frac{(20265\,Pa)(0,1)m^3}{8,31\frac{J}{molK}(358,15K)}$$

$n \approx 0,6808$ moles

b) $V_a$ Se $T = const.$

$$\frac{P_1 V_1}{\cancel{T_1}} = \frac{P_2 V_2}{\cancel{T_2}} \quad P_a V_a = P_b V_b$$

$$\cancel{P_a = \frac{P_b V_b}{V_a}} \quad V_a = \frac{P_b V_b}{P_a}$$

$$V_a = \frac{(0{,}2\,atm) \cdot 0{,}1\,m^3}{0{,}6\,atm} = 0{,}0333\,m^3 = 3{,}3 \times 10^{-2}\,m^3$$

c) $W_{a \to b}$ $\quad W = \int_{V_i}^{V_f} P\,dV$

$$P = \frac{nRT}{V} \Rightarrow W = \int_{V_i}^{V_f} \frac{nRT}{V}\,dV = \ln\left(\frac{V_f}{V_i}\right) nRT$$

$n = 0{,}6808$ moles

$T = 358{,}15\,K$

$$W = (0{,}6808\,mol)\left(8{,}31\,\frac{J}{mol \cdot K}\right) \cdot (358{,}15\,K) \ln\left(\frac{0{,}1\,m^3}{0{,}033\,m^3}\right)$$

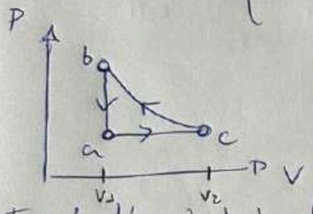
$$\approx 2246{,}34\,J \approx 2{,}25\,kJ$$

d) Pare Gases Ideais

A Energia Interna depende apenas da Temperatura. Como $\Delta T = 0$

$$\Rightarrow \Delta U = 0$$

Questão ③ $\begin{cases} 3 \text{ mols} \\ C_p = 29{,}1 \frac{J}{mol \cdot K} \end{cases}$ $\begin{cases} T_a = 300K \\ T_b = 600K \\ T_c = 492K \end{cases}$




ac: Isobárico
ba: Isocórico
cb: Adiabático.

Trabalho total do ciclo.

$W_{b \to a} = 0$, pois $W = \int P dV = P\Delta V = P(V_1 - V_1) = 0$ //

$W_{a \to c} = \int P dV = P\Delta V = P(V_f - V_1)$

$PV = nRT$ $\quad \cancel{\frac{P_b V_1}{T_b} = \frac{P_a V_1}{T_a}} \quad \cancel{\frac{P_b}{T_b} = \frac{P_a}{T_a}}$

$\cancel{\frac{P_a V_1}{T_a} = \frac{P_a V_2}{T_c}} \quad \cancel{\frac{V_1}{T_a} = \frac{V_2}{T_c}}$

Se $P = $ Const. $\quad P\Delta V = \Delta(PV)$

$\Rightarrow W_{a \to c} = \Delta(PV) = nR\Delta T$

$= 3 mol \cdot 8{,}31 \frac{J}{mol \cdot K} (492 - 300) K$

$W_{a \to c} = 4786{,}56 J$ //

$W_{c \to b}$ No processo Isobárico: $PV^{\gamma} = $ Const

$TV^{(\gamma - 1)} = $ Const. $\quad W = \int P dV$

$$\Delta U = n C_v \Delta T$$

$$C_v = C_p - R = 29{,}1 \frac{J}{mol K} - 8{,}31 \frac{J}{mol K}$$

$$C_v = 20{,}79 \frac{J}{mol \cdot K}$$

$$\Delta U = Q - W$$

No processo Adiabático:

$$Q = 0$$

$$\Rightarrow \Delta U = -W \Rightarrow -W = n C_v \Delta T$$

$$-W = 3 mol \left( 20{,}79 \frac{J}{mol \cdot K} \cdot (300K - 492K) \right)$$

$$W_{c \to b} = -6735{,}96 \ J$$

$$W_{Total} = W_{a \to c} + W_{c \to b} + W_{b \to a}$$

$$= 0 + 4{,}79 \ kJ - 6{,}73 \ kJ$$

$$= -1949{,}4 \ J \cong -1{,}95 \ kJ$$

Questão 4

a) Ciclo de Otto:

1- Compressão Adiabática

2- Aquecimento a ~~volume~~ volume constante.

3- Expansão Adiabática

4- Resfriamento a Volume Constante.

Durante a compressão adiabática:

$$PV^{\gamma} = \text{Const} \qquad TV^{(\gamma-1)} = \text{Const}$$

$$T_1 V_1^{(\gamma-1)} = T_2 V_2^{(\gamma-1)}$$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{(\gamma-1)} \qquad r = \frac{V_2}{V_1} \text{ é a taxa de compressão.}$$

$$\Rightarrow \frac{T_1}{T_2} = r^{\gamma-1} \qquad \eta = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H}$$

$$\eta = 1 - \frac{1}{r^{(\gamma-1)}} \qquad \begin{cases} \gamma = 1{,}40 \\ r = 9{,}50 \end{cases}$$

$$\gamma - 1 = 0{,}4$$

$$r^{(\gamma-1)} = 9{,}5^{0{,}4} \approx 2{,}46 \Rightarrow \eta = 1 - \frac{1}{2{,}46}$$

$$\eta = 0{,}593$$

b) Se o eficiência é 0,593

$\cancel{Q_H + Q_C = W}$ $\quad Q_H + Q_C = W$

$Q_H = 10000\ J = 10\ kJ$

$W = \eta Q_H \qquad W = 5,93\ kJ$

$Q_C = 5,93\ kJ - 10\ kJ$

$= -4,06\ kJ$

$|Q_C| = 4,06\ kJ$

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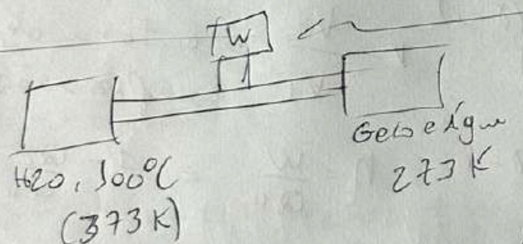
(5)




$Q = mL$

$L_f = 3,34 \times 10^5\ \frac{J}{kg}$

$|Q_H| - |Q_C| = |W| \qquad Q_C = mL_f$

$= 0,04\ kg \cdot 3,34 \times 10^5\ \frac{J}{kg}$

$= -13,360\ kJ$

$\frac{Q_H}{T_H} = -\frac{Q_C}{T_C} \qquad Q_H = Q_C \left(\frac{-T_H}{T_C}\right)$

$Q_H = -13,360\ kJ \left(-\frac{373\ K}{273\ K}\right) = 18,23\ kJ$

$|Q_H| + |Q_C| = W = 18,23\ kJ - 13,360\ kJ$

$= 4,892\ kJ$ //